

Lab Experiment: 3

**Title: Inspection of Voltage Variation with Light Dependent Resistor (LDR)
using Arduino (Analog Sensor Interfacing)**

Course title: IOT Laboratory

*Course code: CSE-460
4th Year 1st Semester Examination 2025*

Date of Submission: 22/02/2026



Submitted to-

Dr. Md. Ezharul Islam, B.SC.(JU), M.ENG.G.(JAPAN), PHD(JAPAN)

Professor

*Department of Computer Science and Engineering
Jahangirnagar University*

&

Samsun Nahar Khandakar

Assistant Professor

*Department of Computer Science and Engineering
Jahangirnagar University
Savar, Dhaka-1342*

Class Roll	Exam Roll	Name
383	210903	Abdullah Nazmus-Sakib
379	210899	Shah Sultan
391	210911	Md Khaled Mahmud

1. Title Inspection of Voltage Variation with Light Dependent Resistor (LDR) using Arduino

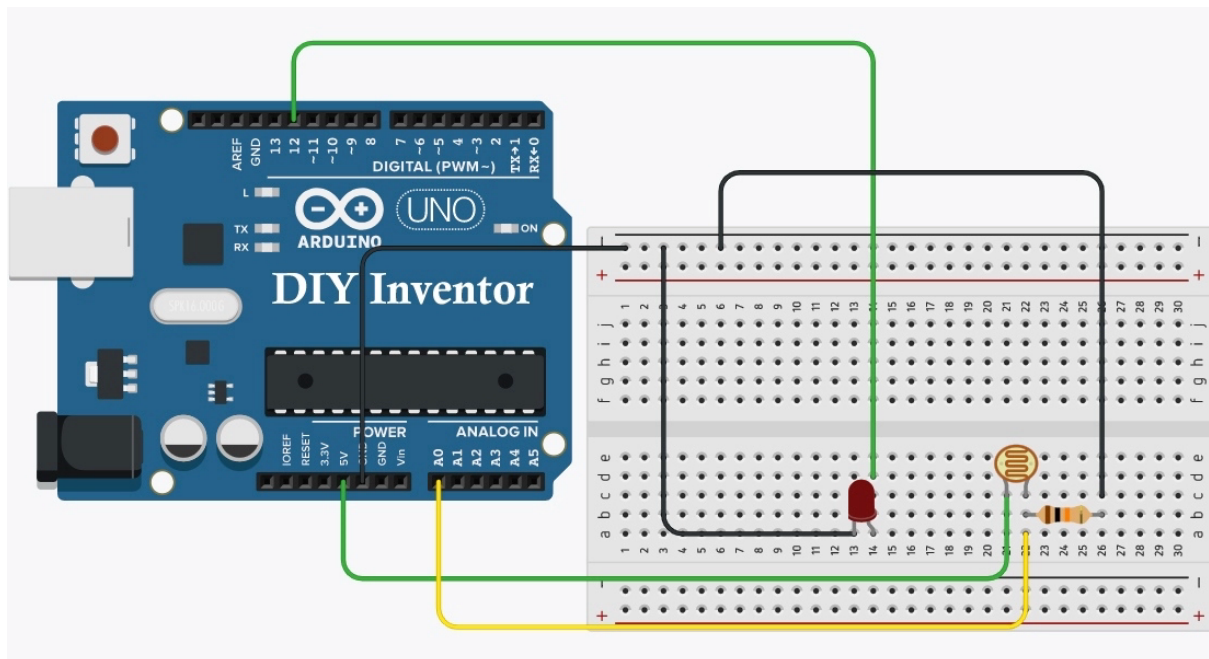
2. Objective

- To interface an analog sensor (LDR) with Arduino
- To observe voltage variation with changing light intensity
- To convert analog input into digital output using ADC
- To control LED brightness based on light level

3. Components Required

COMPONENT	QUANTITY
ARDUINO UNO	1
LDR (LIGHT DEPENDENT RESISTOR)	1
LED	1
RESISTOR (10 KΩ FOR LDR)	1
RESISTOR (220–330 Ω FOR LED)	1
JUMPER WIRES	As required
BREADBOARD	1
USB CABLE	1

4. Circuit Connection



◆ LDR Voltage Divider

- One terminal of LDR → **5V**
- Other terminal of LDR → **Analog Pin A0**
- 10 k Ω resistor from A0 → **GND**

👉 This creates a voltage divider whose output depends on light intensity.

◆ LED Connection

- LED Anode (+) → **Digital Pin 9 (PWM)**
- LED Cathode (–) → **GND through 220–330 Ω resistor**

🖨️ 5. Program (Arduino Code)

```
const int analogInPin = A0;    // LDR input
const int analogOutPin = 9;    // LED output (PWM)

int sensorValue = 0;
int outputValue = 0;

void setup() {
  Serial.begin(9600);
}

void loop() {
  sensorValue = analogRead(analogInPin);
  outputValue = map(sensorValue, 0, 1023, 0, 255);
  analogWrite(analogOutPin, outputValue);

  Serial.print("sensor = ");
  Serial.print(sensorValue);
  Serial.print("\t output = ");
  Serial.println(outputValue);

  delay(1000);
}
```

⚙️ 6. Working Principle

- LDR resistance changes with light intensity
 - ☀️ More light → Lower resistance
 - ● Less light → Higher resistance
- The voltage divider converts resistance change into voltage change
- Arduino reads this voltage via **Analog-to-Digital Converter (ADC)**
- ADC converts voltage into values from **0–1023**
- These values are mapped to **0–255** for PWM output

- PWM signal controls LED brightness

👉 Result: LED brightness varies according to surrounding light.

7. Procedure

1. Collect all required components
2. Build the circuit according to the diagram
3. Connect Arduino to computer via USB
4. Open Arduino IDE
5. Write or paste the code
6. Select correct Board and Port
7. Upload the program
8. Open Serial Monitor to observe sensor readings
9. Change light conditions (cover/uncover LDR) and observe LED brightness

8. Observation

- When light intensity increased → Sensor value changed → LED brightness changed
- Serial Monitor displayed corresponding analog readings
- System responded smoothly to light variation

9. Result

The experiment successfully demonstrated voltage variation with light intensity using an LDR and analog interfacing with Arduino. LED brightness was controlled according to ambient light conditions.

10. Applications

- Automatic street lighting systems
- Smart home lighting
- Light meters
- Solar tracking systems
- Security systems
- IoT environmental monitoring

11. Precautions

- Use correct resistor values
- Ensure proper polarity of LED

- Avoid loose connections
- Do not short 5V to GND
- Check analog pin connection carefully

🏁 12. Conclusion

This experiment provided practical understanding of analog sensor interfacing in IoT systems. It demonstrated how environmental light can be measured using an LDR and used to control devices automatically. The knowledge gained is fundamental for building smart automation and sensing systems.